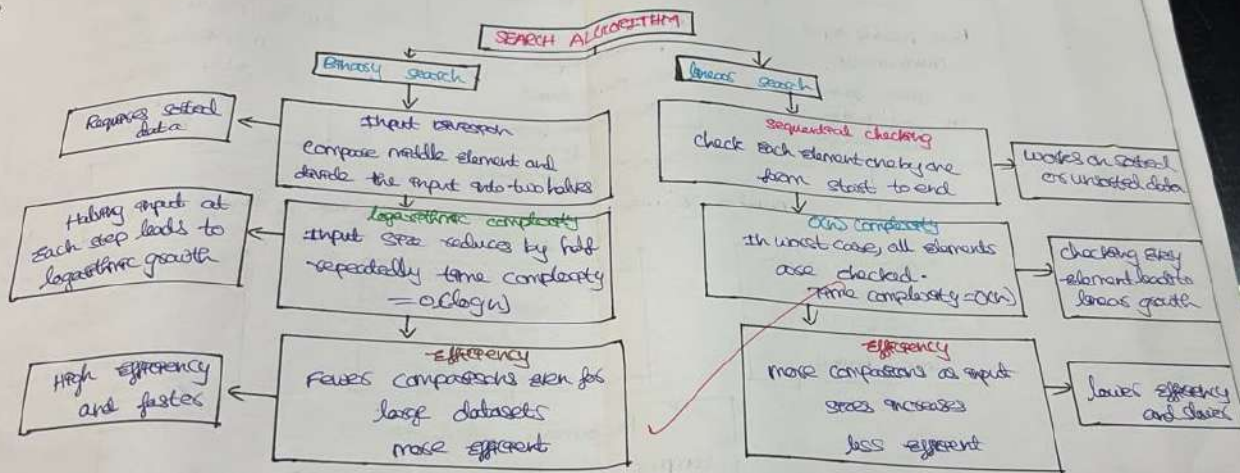
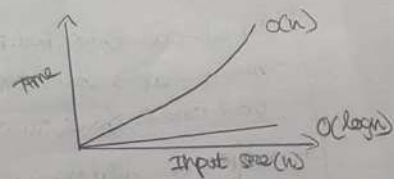


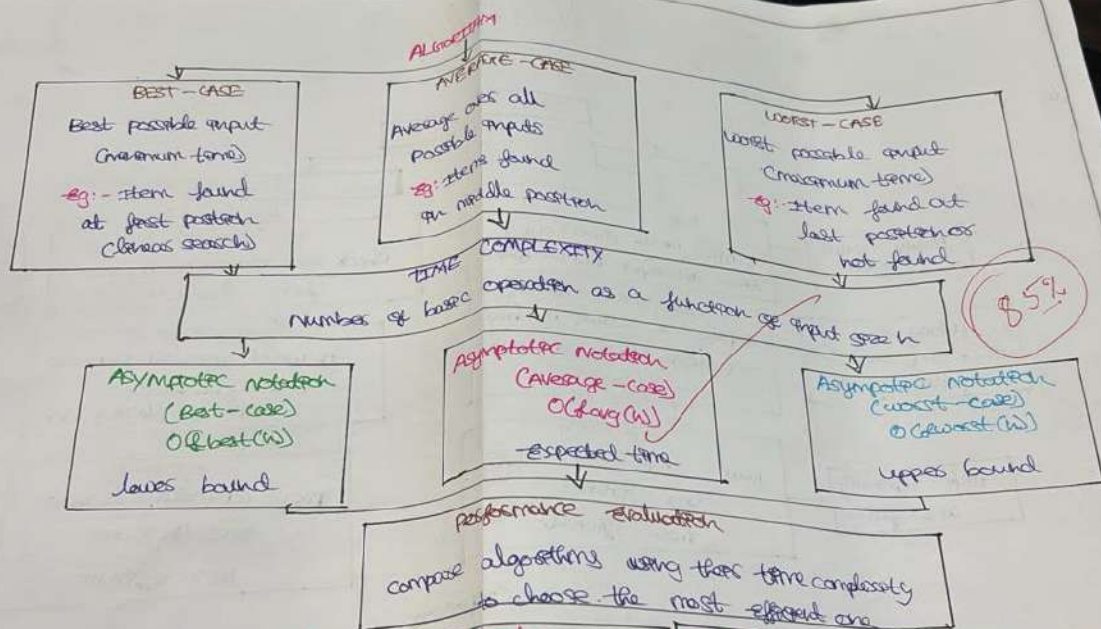
CONCEPT MAP

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- Comparison
- Binary search divides input in half each time → $O(\log n)$
 - Linear search checks one by one → $O(n)$
 - As n increases, $\log n$ grows much slower than n
 - Therefore, Binary search has a superior growth rate





Why Analyze all three cases?

- Best-case shows best performance
- Average-case → shows typical behavior
- Worst-case → shows maximum time required

In short

Algorithm behavior varies with input. All three cases give a complete

How this map helps in real-time comparison

- provides complete view under different inputs.
- Considers worst-case for reliability and safety
- Makes comparison more practical